1. Finding the Complementary Function (y_c)

When solving higher-order linear differential equations of the form $f(D)y = F(x)$, the general solution is always composed of two parts:

$$y = y_c + y_p$$

The **Complementary Function** (y_c) represents the solution to the **homogeneous** version of the equation (where the right side is zero). This is the first step in solving *any* linear ODE with constant coefficients.

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1. The Operator 'D'

To simplify writing derivatives, we use the differential operator D: * $Dy = \frac{dy}{dx}$ * $D^2y = \frac{d^2y}{dx^2}$ * $D^ny = \frac{d^ny}{dx^n}$

An equation like $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0$ can be written as:

$$(D^2 + 5D + 6)y = 0$$

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2. The Auxiliary Equation (A.E.)

To find y_c, we assume a solution of the form $y = e^{mx}$. Substituting this into the homogeneous equation gives us the **Auxiliary Equation** (or Characteristic Equation).

Simply replace D with m and set the equation to zero:

$$f(m) = 0$$

We then solve for the roots of m. The nature of these roots determines the form of y_c.

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3. Cases for Roots and Forms of y_c

Case 1: Real and Distinct Roots

If the roots $m_1, m_2, ...$ are real and different:

$$y_c = C_1e^{m_1x} + C_2e^{m_2x} + ...$$

Case 2: Real and Repeated Roots

If a root m is repeated k times (e.g., m, m, m):

$$y_c = (C_1 + C_2x + C_3x^2 + \cdots + C_kx^{k-1})e^{mx}$$

Case 3: Complex Conjugate Roots

If the roots are complex ($\alpha \pm i\beta$):

$$y_c = e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$$

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4. Solved Questions

Here are examples from the lecture notes demonstrating how to find y_c for different types of roots.

Question 1: Real & Distinct Roots

Find y_c for: $y'' + 6y' + 5y = 0$

Solution: 1. **Write in Operator Form:** $(D^2 + 6D + 5)y = 0$ 2. **Auxiliary Equation:** $m^2 + 6m + 5 = 0$ 3. **Solve for Roots:** Factor the quadratic: $(m + 5)(m + 1) = 0$ Roots: $m_1 = -1, m_2 = -5$ 4. **Write y_c:** Since roots are real and distinct:

$$y_c = C_1 e^{-x} + C_2 e^{-5x}$$

Question 2: Real & Repeated Roots

Find y_c for: $y''' + 3y'' + 3y' + y = 0$

Solution: 1. **Write in Operator Form:** $(D^3 + 3D^2 + 3D + 1)y = 0$ 2. **Auxiliary Equation:** $m^3 + 3m^2 + 3m + 1 = 0$ 3. **Solve for Roots:** Recognize this as the binomial expansion of $(m+1)^3$: $(m+1)^3 = 0$ Roots: $m = -1, -1, -1$ (Repeated 3 times) 4. **Write y_c:** Since the root -1 repeats 3 times, we multiply the constants by increasing powers of x:

$$y_c = (C_1 + C_2 x + C_3 x^2)e^{-x}$$

Question 3: Complex Roots

Find y_c for: $y'''' + 4y = 0$

Solution: 1. **Write in Operator Form:** $(D^4 + 4)y = 0$ 2. **Auxiliary Equation:** $m^4 + 4 = 0 \implies m^4 = -4$ $m^2 = \pm 2i$ $m = \pm\sqrt{2i}$

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*Alternative Method (Factorization):*
$m^4 + 4 = (m^2 + 2)^2 - 4m^2 = (m^2 + 2 - 2m)(m^2 + 2 + 2m) = 0$

Solve $m^2 - 2m + 2 = 0$ using quadratic formula:
$m = \frac{2 \pm \sqrt{4 - 8}}{2} = \frac{2 \pm 2i}{2} = 1 \pm i$

Solve $m^2 + 2m + 2 = 0$:
$m = \frac{-2 \pm \sqrt{4 - 8}}{2} = -1 \pm i$

Roots: $1 \pm i$ and $-1 \pm i$
($\alpha = 1, \beta = 1$) and ($\alpha = -1, \beta = 1$)
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3. **Write y_c:** Combine the solutions for both pairs of conjugate roots:

$$y_c = e^{x}(C_1 \cos x + C_2 \sin x) + e^{-x}(C_3 \cos x + C_4 \sin x)$$

Next Step: Once y_c is found, we must find the Particular Integral (y_p) based on the function on the right side of the equation. Proceed to Topic 2: Particular Integral - Exponential Case.